

3 Testing Report

3.1 Testing Rationale and Objectives

The usability test assessed whether the prototype allowed participants to complete the core tasks without help. It checked each acceptance criterion and showed where users misunderstood a control, a message, or a change in system state. The team compared observed behaviour with completion data and participant comments. The findings were then used to decide which requirements and screens needed revision before the next iteration.

3.2 Participants and Method

3.2.1 Participants

Five purposively selected participants covered student and working-adult contexts, different rental and shared-living experience, colour-accessibility, mobile-first use and different English-language backgrounds (Table 1). They were recruited through the project team network and screened for these contexts. Participation was voluntary and unpaid, and consent was obtained before each session.

Table 1. Iteration 1 participant profile.

ID	User group	Relevant context
User A	UNSW student	High cleanliness standards and interest in household hygiene.
User B	Domestic UNSW student	Colour vision deficiency.
User C	Employed UNSW graduate living in a shared apartment	English is not the first language.
User D	Newly arrived international student	No Australian rental experience; primarily uses a mobile phone.
User E	Young professional	More than three years of shared-living experience; previously searched for housemates and established common-area rules.

3.2.2 Procedure and Measures

Five participants tested version 1.3 of the HTML prototype across one tablet, two laptops and two mobile devices between 25 and 27 July 2026. The allocation is shown in Table 2.

Table 2. Iteration 1 participant device allocation.

User	Date	Device	Browser	Viewport
User A	25 July 2026	iPad tenth generation	Safari	1180 by 820
User B	25 July 2026	MacBook Air M2	Chrome	1440 by 900
User C	26 July 2026	Dell Inspiron 14	Microsoft Edge	1366 by 768
User D	26 July 2026	iPhone 13	Safari	390 by 844
User E	27 July 2026	Samsung Galaxy S22	Chrome	360 by 800

Each moderated session used the same seeded account and task state. Participants attempted T1 to T5 while thinking aloud. The facilitator gave a hint only after a participant was unable to continue for 30 seconds. The observer recorded final-state completion, time, errors, assistance and interpretation of feedback. Wrong controls, pages or unsupported actions counted as errors; assisted

completions were not independent. Only task means were available, so individual time-threshold compliance could not be confirmed and will be captured in Iteration 2.

3.3 Iteration 1 Testing

3.3.1 Test Plan and Acceptance Criteria

Iteration 1 retained the five tasks and acceptance-criteria identifiers from Part 2, but expanded coverage from the original mobile viewport to tablet, laptop and mobile contexts. A task passed when at least four of five participants completed it independently. AC-US11-01 and AC-US09-01 used seeded, simulated states, so testing assessed usability rather than production verification or messaging security.

Table 3. Iteration 1 test plan and acceptance criteria coverage.

Test	User task	Traceability	Acceptance criteria	Measurable success threshold
T1	Narrow properties using combined search criteria	US-13; S-01	AC-US13-01 to AC-US13-04 and AC-US13-06 (implemented)	At least 4/5 independently complete the filtered search, explain affordability, recover from no results and use a custom address within three minutes.
T2	Recognise verified housemate profiles	US-11; S-02	AC-US11-01 (simulated) and AC-US11-02 (implemented)	At least 4/5 independently find the verified-only control, identify its cues and explain that verification is not a guarantee within 90 seconds.
T3	Build a profile and use recalculated compatibility	US-12; S-03	AC-US12-01, AC-US12-02 and AC-US12-04 to AC-US12-06 (implemented)	At least 4/5 independently complete the profile, explain the top match and recover from validation and zero-result states within three minutes.
T4	Complete phone verification and message safely	US-09; S-04	AC-US09-01 (simulated) and AC-US09-04 (implemented)	At least 4/5 independently pass verification and message within three minutes; contact details remain blocked with a clear explanation.
T5	Propose, review and activate a house rule	US-02, US-10; S-05, S-06	AC-US02-01 to AC-US02-03, AC-US10-01, AC-US10-02 and AC-US10-04 (implemented)	At least 4/5 independently view, propose and activate a rule, including language, validation and approval states, within four minutes.

3.3.2 Testing Outcomes

Table 4. Iteration 1 outcome summary.

Test	Result	Quantitative outcome	Acceptance-criteria assessment and consequence
T1	Pass	5/5 reached the final state; 4/5 independently; mean 2 min 8 sec; 2 errors; 1 assisted	AC-US13-01 to AC-US13-04 met the threshold; empty-state recovery remained unclear. AC-US13-06 was not separately recorded. Make the reset action more prominent and explicit.
T2	Pass	5/5 reached the final state independently; mean 1 min 2 sec; 1 error; no assistance	Simulated AC-US11-01 and implemented AC-US11-02 met the threshold. Two users found the explanation too small; enlarge it and retain non-colour cues.
T3	Partial	4/5 reached the final state; 3/5 independently; mean 2 min 44 sec; 3 errors; 1 assisted	AC-US12-01 was demonstrated; AC-US12-02 was partial because the score was unclear. AC-US12-04 to AC-US12-06 require criterion-level retesting. Enlarge the factor explanation and show why the score changed.
T4	Partial	5/5 reached the final state; 4/5 independently; mean 2 min 18 sec; 2 errors; 1 assisted	Simulated AC-US09-01 met the threshold. AC-US09-04 blocked contact details, but its reason and next step were unclear. Link the warning to the contact-sharing action.
T5	Partial	4/5 reached the final state; 2/5 independently; mean 3 min 11 sec; 4 errors; 2 assisted	AC-US10-01 was demonstrated; AC-US10-02 was partial because approval progress was unclear. US-02 and AC-US10-04 require criterion-level retesting. Combine and emphasise approval and activation status.

All five reached the final state in T1, T2 and T4, but only T2 required no assistance. T3 and T5 had the lowest independent completion and most errors. Problems clustered around post-action feedback: empty-search recovery, score meaning, contact blocking and rule approval. Iteration 2 will keep participant-level times and separate results for every criterion.

3.3.3 User Feedback

Table 5 reports the seven accepted participant-level findings. Device and viewport allocations are given in Table 2. The statements paraphrase observation notes rather than claiming verbatim participant quotations.

Table 5. Participant-level accessibility and device-specific findings.

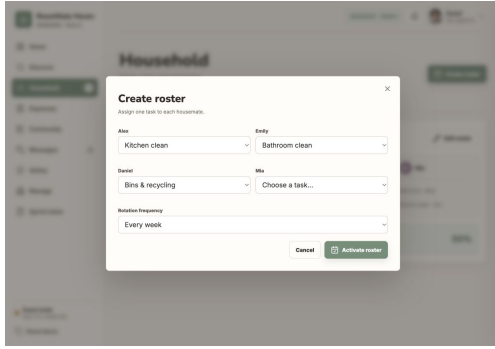
User	Task	Observed issue	Effect on task
A	T5	Keyboard focus moved outside the Create roster dialog. Escape did not close it, and focus was not returned to the opening control after closing.	The participant lost the interaction context and required assistance.
B	T2	The Discover tabs did not respond to the arrow keys. Their selected state also relied mainly on colour and a thin underline.	The participant completed the task with a pointer, but keyboard navigation and the selected state were unclear.
C	T4	The privacy warning appeared after contact details were entered, but its relationship to the icon-only contact-sharing action was unclear.	The participant paused for more than 30 seconds and continued after one facilitator hint.
C	T5	Selecting Chinese translated rule content while surrounding headings, labels and actions remained in English.	The mixed-language state reduced clarity and made the language setting appear incomplete.

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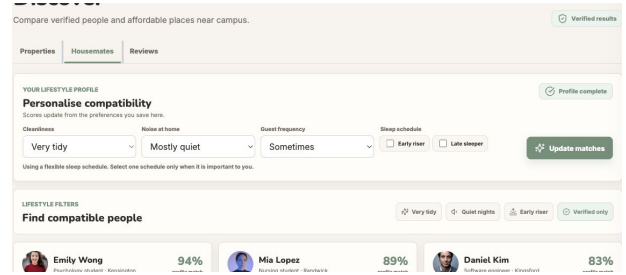
Table 5. Participant-level accessibility and device-specific findings (continued).

User	Task	Observed issue	Effect on task
D	T4	At 390 by 844 pixels, the conversation list and New message control were hidden with no visible mobile replacement.	This restricted mobile conversation management because the participant could not select Daniel or begin a new conversation.
E	T5	Selecting Save draft produced no visible confirmation, navigation or control-state change.	The participant selected the control twice and assumed it was not working.
E	T5	Approval progress was displayed separately from the activation condition and was not sufficiently prominent.	The participant requested clarification and did not complete the task independently.

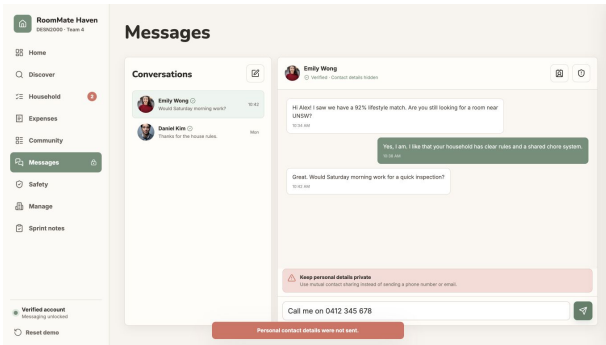
The recurring problems concerned keyboard interaction, non-colour state cues, language consistency, mobile access and post-action feedback. The corresponding recreated states are embedded in Figures 1–2; a fuller description of each image is stored in `tests/participant-findings.md`. Screenshots document interface states, while pauses, assistance and participant interpretation come from the observation notes in Table 5. The roster-dialog, conversation-switching and Save draft observations were supplementary checks made during the named task sessions; they were not used to revise the aggregate results in Table 4.



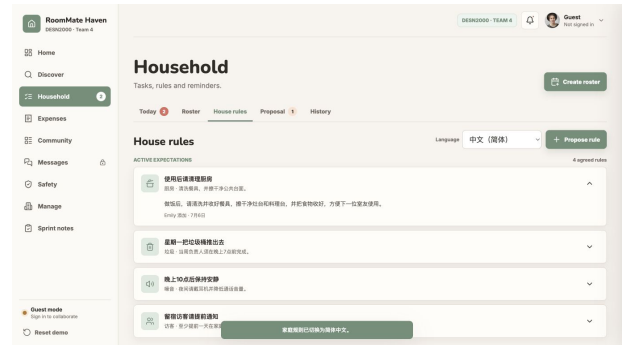
(a) User A/T5: roster-dialog state. Focus escape, Escape-key behaviour and focus return were verified through interaction checks.



(b) User B/T2: Discover selected-state styling. Arrow-key behaviour was verified through an interaction check.

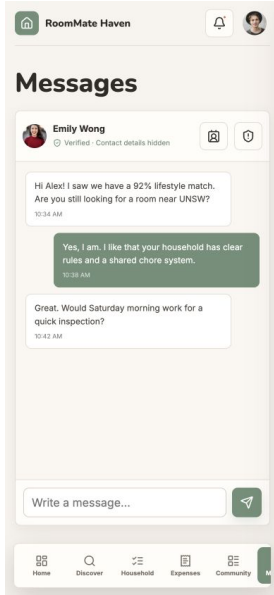


(c) User C/T4: blocked contact details, privacy warning and the separate icon-only contact-sharing control.

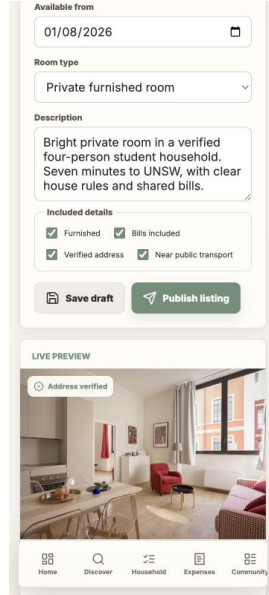


(d) User C/T5: translated rule content surrounded by English headings, labels and actions.

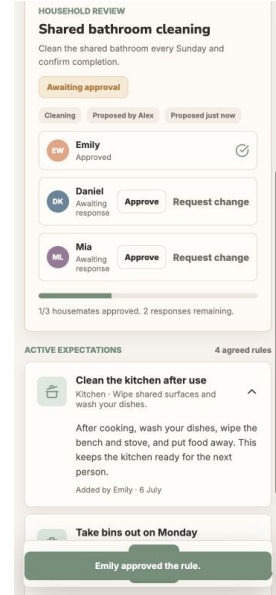
Figure 1. Tablet and laptop evidence for the participant-level findings in Table 5.



(a) User D/T4: mobile message view without a visible conversation switcher or New message control.



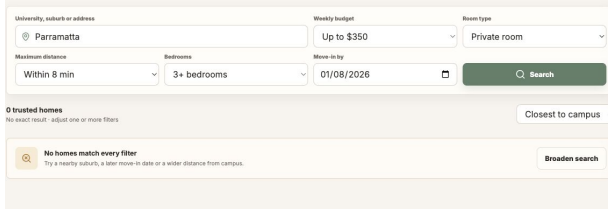
(b) User E/T5: Save draft after activation, with no visible confirmation or state change.



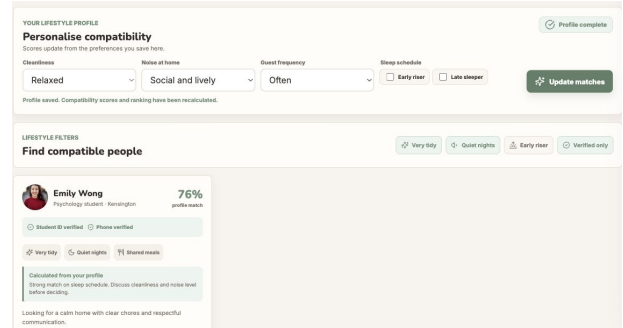
(c) User E/T5: one-of-three approval state and the separate activation-status information.

Figure 2. Mobile evidence for the participant-level findings in Table 5.

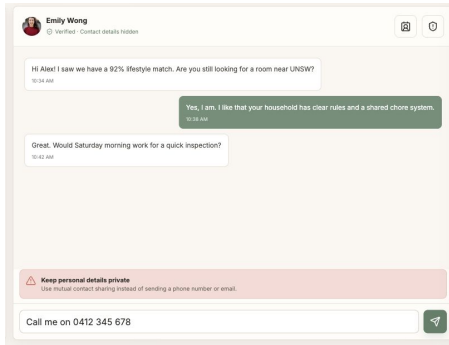
Figure 3 consolidates the original task-level post-session recreations captured from the same Sprint 1 HTML build at a 1440 by 900 viewport using local Playwright on 27 July 2026. They document the interface states before Iteration 2 refinement, rather than participant behaviour; the behavioural evidence is reported in Table 4.



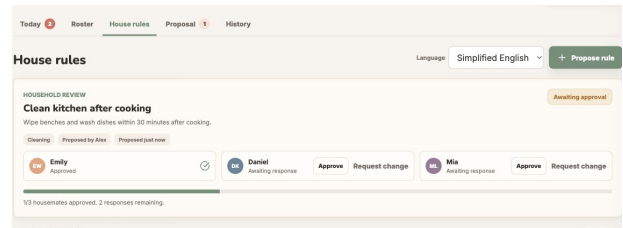
(a) T1, AC-US13-04: the secondary “Broaden search” action supported a more prominent reset action.



(b) T2-T3, AC-US11-02 and AC-US12-02: small trust and score explanations supported larger text and non-colour cues.



(c) T4, AC-US09-04: the warning and icon-only sharing control were separated, making the next step less apparent.



(d) T5, AC-US10-02: separated approval and activation indicators supported a stronger combined status message.

Figure 3. Compact Sprint 1 task-level baseline evidence before Iteration 2 refinement.

3.4 Changes after Sprint 1

The team changed states that failed, remained partial or blocked keyboard and mobile use in Iteration 1. Table 6 links each design change to its evidence and revised Jira issue. The Jira cells remain blank until the revised issues are published.

Table 6. Design changes made after Sprint 1.

Task	Design modification	Evidence	Updated Jira link
T1	The empty state places Broaden search beside guidance that explains how to recover.	Figure 4a	
T2	Verification cues pair an icon with readable status text.	Figure 4b	
T2	Tabs expose their selected state and respond to the arrow, Home and End keys.	Figure 5	
T3	Each match card explains how the saved profile affected its score and names a point to discuss.	Figure 4c	
T4	The message warning blocks the content, explains mutual sharing and announces the result through an alert.	Figure 6a	
T4	The mobile message view keeps a conversation selector and a New control.	Figure 6b	
T5	Open dialogs keep keyboard focus, close with Escape and return focus to the opening control.	Figure 7a	
T5	Roster validation links the error to the invalid field and moves focus to it.	Figure 7a	
T5	Chinese rule content exposes the zh-CN language code to assistive technology.	Figure 7b	
T5	Saving a draft updates the page status and shows a confirmation.	Figure 7c	

3.5 Iteration 2 Test

Iteration 2 repeated the affected Sprint 1 tasks with the same five participants after the prototype was revised. The results below report their new testing outcomes and are supported by the implemented states shown in Figures 4 to 7.

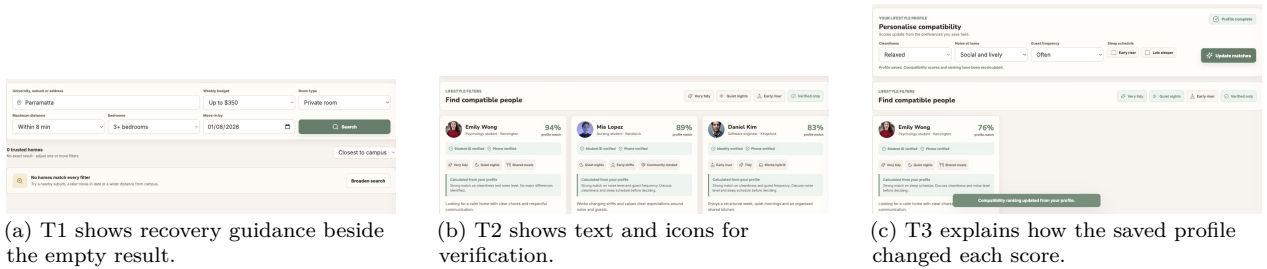


Figure 4. Iteration 2 implementation evidence for T1 to T3.

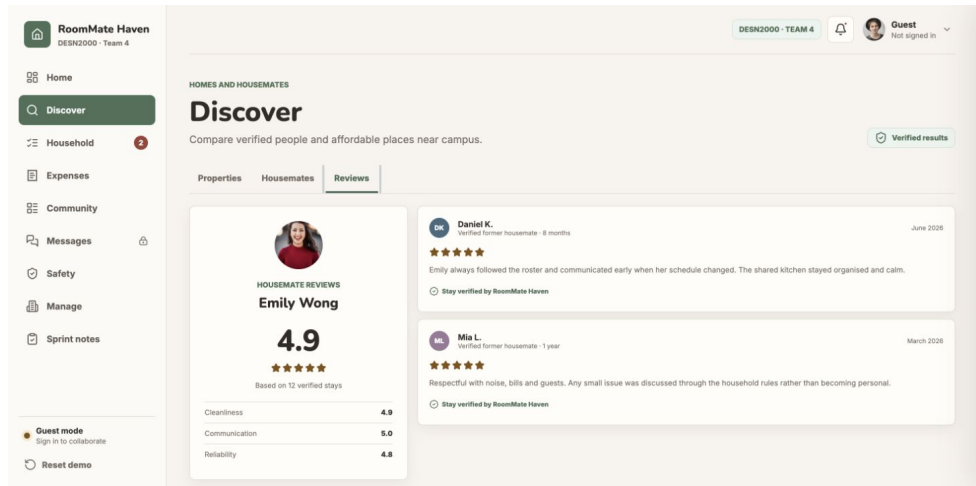
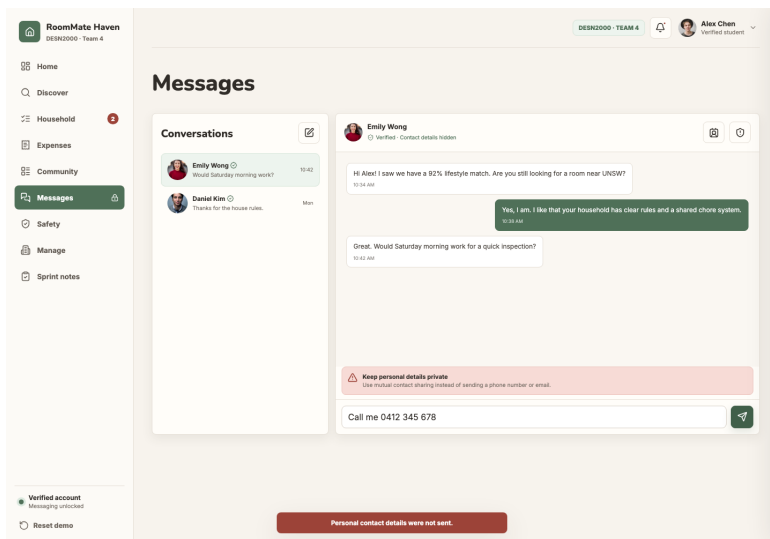
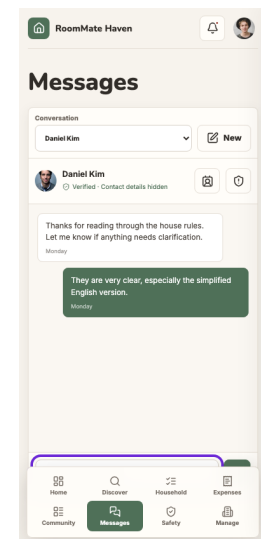


Figure 5. T2 tab focus and selected state after keyboard navigation.

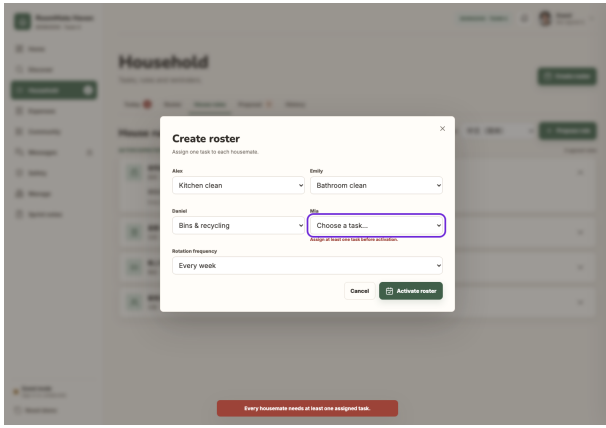


(a) T4 blocks a phone number and shows the safer next step.

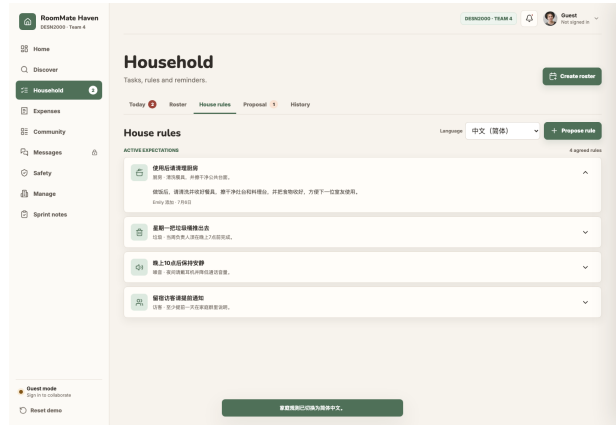


(b) T4 keeps conversation controls at the mobile viewport.

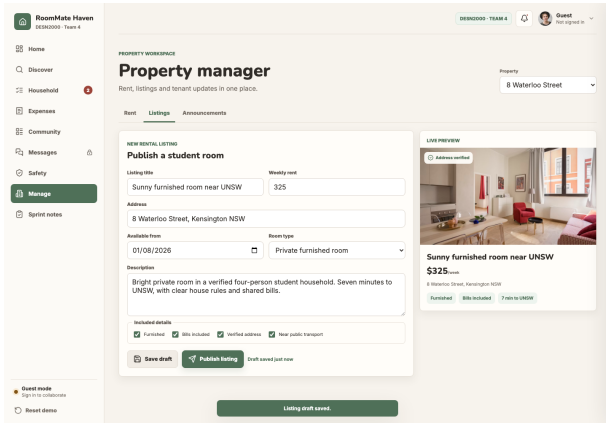
Figure 6. Iteration 2 implementation evidence for T4.



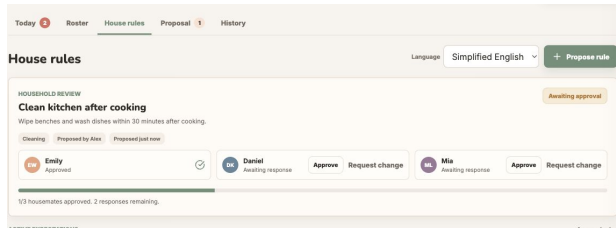
(a) T5 keeps focus on the invalid roster field.



(b) T5 displays Chinese rules with matching language metadata.



(c) The supplementary draft check shows a saved state.



(d) T5 still separates progress from the activation condition.

Figure 7. Iteration 2 implementation evidence for T5 and its remaining issue.

Table 7. Iteration 2 results for the modified Sprint 1 tasks.

Task	Resolved problem	Acceptance criteria result	New issue	Remaining problem
T1	The empty result gives a recovery action and names the inputs that can change.	AC-US13-04 changed from Fail to Pass.	No.	None shown.
T2	Text supports each verification cue. Keyboard users can identify and change tabs.	AC-US11-02 remains Pass. No failed criterion changed to Pass.	No.	Participants still confused profile identity verification with the phone check that unlocks messaging.
T3	The match card explains the score source and identifies points to discuss.	AC-US12-02 remained Partial because several participants still treated the score as a safety signal.	No.	The score source is visible, but its limit remains unclear.
T4	Contact details remain unsent. The warning explains mutual sharing and the mobile view retains both conversation actions.	AC-US09-04 changed from Fail to Pass.	No.	The desktop sharing action still relies on an icon, so its purpose is less clear without hover.
T5	Dialog focus, roster errors, rule language metadata and draft feedback expose a clear state.	AC-US02-02 remains Pass. No failed criterion changed to Pass.	No.	AC-US10-02 remains Fail because the status does not explain activation. The Household entry and House rules destination use different labels. Surrounding controls remain in English after Chinese is selected.

3.6 Persistent Issues and Impact

The same five participants tested both iterations. Several still struggled to explain what the compatibility score measured after the second test. Although the revised cards linked the score to saved preferences, some users treated a high score as evidence that a person was safe. This could cause them to misjudge a potential housemate.

Confusion also persisted between a verified profile and phone verification. Similar visual cues made the two states appear equivalent, even though identity verification supports profile trust and phone verification only unlocks messaging. The ambiguity increased decision time and weakened confidence in the safety process.

Navigation remained unclear because the main entry was named Household while the destination was named House rules. Users opened the wrong view before finding rule proposals. They also struggled in both rounds to connect approval progress with activation. These problems caused repeated checking and left users uncertain about which household rules applied.